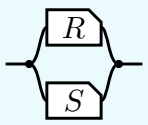
Relation Algebra

§4. ∩ is a commutative idempotent monoid on every hom-set

(4a)

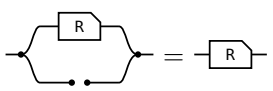
The **meet** of `R,S : a→b`, the paper's **convolution**, is `R∩S:=◁(R⊗S)▷` — copy the input, run `R` and `S` on the two copies, merge the results — so what comes out is what both of them do.



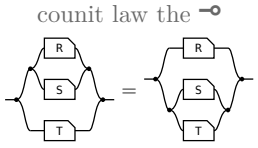
transcribed: a definition has no statement to export

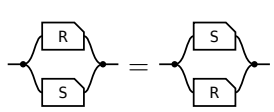
On every hom-set it is associative, commutative and idempotent , , with unit the maximal arrow `⊤=⊸∘⊸` (the paper's Lemma 4.11).

(4b)

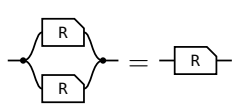


unit: one half of `⊤` per end — the merge's unit law absorbs the `⊸`, the copy's counit law the `⊸`





commutative: `σ` crosses `R⊗S` by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; `⊗` re-brackets for nothing, being strict here

idempotent: the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So `≤` is the order this monoid induces. `R∩S≤R` comes from the unit, and idempotency turns anything under both `S` and `T` into something under `S∩T`, since `R=R∩R≤S∩T`.

And one law relating `∩` to composition, which is **not** an equation:

(4c)

semi-distributivity, and what supplies it	**picture**
`R (S∩T)⊑RS∩RT` — the lax copy law. Equality exactly when `R` is single valued: the Maps section's `F(R∩S)=FR∩FS`.	